INSIGHT PAPER

AI Governance for Critical Infrastructure

Why traditional governance models break when applied to AI-augmented platforms, and what to do about it.

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Situation. Complication. Resolution.

> **Critical infrastructure operators are deploying AI across their platforms—yet the governance models they rely on were designed for a pre-AI world. The gap is not theoretical. It is costing time, trust, and regulatory credibility.**

Situation. Telecommunications, energy, transport, and financial infrastructure operators are under pressure to adopt AI—for network optimization, predictive maintenance, customer operations, and security. The business case is proven. The technology is mature. But the governance frameworks that govern these platforms have not evolved at the same pace.

Complication. Traditional governance—change control boards, release gates, manual compliance checks—assumes deterministic systems with known failure modes. AI models are probabilistic. They degrade, drift, and produce outputs that cannot be predicted at design time. When a model misclassifies a network fault, or a recommendation engine steers an operator toward a suboptimal decision, who is accountable? Which committee reviews the training data lineage? How do you audit a system that learns?

Resolution. AI governance for critical infrastructure requires a new operating model—one that combines platform engineering rigour with AI-native guardrails. Not a parallel bureaucracy. An embedded capability. Govantis has developed this model through years of designing governance for the world's most complex telecom platforms, now adapted for the AI era.

This paper outlines the challenge, the framework, and the path forward.

The Infrastructure Governance Imperative

Critical infrastructure is undergoing its most significant transformation since digitization. AI is no longer experimental—it is being deployed in production across network operations centres, customer platforms, and security monitoring. **88% of organizations now regularly use AI in at least one business function** (McKinsey Global Survey on AI, November 2025).

Yet the same data reveals a stark gap: **over 80% of organizations report no meaningful enterprise-wide EBIT impact from AI** (McKinsey, 2025), and only **5% achieve substantial AI value at scale** (BCG AI Value Survey, 2025). Among the multiple blockers—talent, data quality, unclear ROI—governance is the least addressed. The EU AI Act now imposes fines of up to €35 million or 7% of global turnover for non-compliance, yet most operators lack the governance infrastructure to demonstrate conformity.

88%

ORGANIZATIONS REGULARLY USING AI

80%+

REPORT NO EBIT IMPACT FROM AI

5%

ACHIEVING SUBSTANTIAL AI VALUE AT SCALE

Sources: McKinsey Global Survey on AI, November 2025 · BCG AI Value Survey, October 2025 · EU AI Act Regulation (EU) 2024/1689

THE REGULATORY WAVE

Three converging pressures

1. The EU AI Act. The world's first comprehensive AI regulation classifies infrastructure AI as "high-risk." This imposes requirements for risk management, data governance, transparency, human oversight, and accuracy across the model lifecycle. Non-compliance can result in fines of up to €35 million or 7% of global annual turnover—figures that concentrate minds in legal and compliance departments across every critical infrastructure sector.

2. Sector-specific regulation. Telecom regulators (BEREC, Ofcom, ANACOM) are developing AI-specific guidelines. The intersection of AI with existing telecom regulation (net neutrality, spectrum management, emergency services) creates a compliance matrix that traditional governance cannot manage. Gartner forecasts that AI-specific regulatory instruments will quadruple, covering an estimated 75% of the world's economies by 2030.

3. Standards and frameworks. The NIST AI Risk Management Framework, ISO/IEC 42001 (AI management systems), and the emerging IEEE P7000 series provide guidance but no off-the-shelf governance model. Operators must build their own. As industry analysts note, AI oversight has moved "beyond principles into a discipline requiring centralized inventory, risk management, and continuous monitoring."

> **The governance gap:** Most operators have invested heavily in AI capability (talent, platforms, models) but have not invested proportionally in AI governance. The result is deployment bottlenecks, regulatory exposure, and missed opportunities. BCG found that only 5% of companies achieve substantial AI value at scale (BCG AI Value Survey, 2025), and the primary blockers are organizational — governance, processes, and change management — not technology.

Where Traditional Governance Breaks

Having designed and operated governance programmes for multi-stakeholder telecom platforms over two decades, we have observed six failure patterns that repeat across organizations deploying AI in critical infrastructure.

1. Deterministic Assumptions

Change control expects known outputs. AI produces distributions, not points. Testing "all possible states" is impossible when the model adapts.

2. Accountability Gaps

Who owns an AI decision? The model owner? The platform operator? The vendor? Most RACI matrices assign AI accountability to roles that do not exist yet.

3. Audit Trails That Lie

Traditional logs capture "what happened." AI governance needs to capture "why the model thought that was the right answer"—training data, feature importance, drift metrics.

4. Human-in-the-Loop Theatre

Regulators require human oversight. But a human reviewing 100 AI recommendations per shift is not oversight—it's rubber-stamping. Effective HITL requires purpose-built tooling.

5. Vendor Lock-in by Default

AI governance frameworks from cloud providers tie operators to their ecosystem. A multi-vendor, multi-model infrastructure needs a vendor-agnostic governance layer.

6. Speed vs. Safety False Dichotomy

The assumption that governance slows AI deployment is self-fulfilling when governance is designed as a gate rather than an enabler. The right model accelerates both.

Estimativa Govantis: A pattern we observe consistently across multi-stakeholder programmes is the attempt to apply deterministic change control to probabilistic AI systems. Organizations that treat AI governance as an extension of existing IT governance often face deployment delays measured in months per model while governance boards develop comfort with probabilistic decision-making. The solution is not more process—it is different process.

AI-Native Governance: A Framework

AI-native governance is not a compliance overlay. It is a re-engineering of how governance operates—from static gates to continuous, data-driven guardrails. The Govantis framework is built on three pillars, each corresponding to a phase of the AI model lifecycle.

01

Design-Time Governance

Before a model reaches production, governance must be embedded in the development process. This includes: data lineage and provenance tracking, bias and fairness testing as part of CI/CD, model card creation (documenting intended use, limitations, performance characteristics), and risk classification by domain (network, customer, security, financial).

02

Runtime Governance

In production, governance shifts from gates to guardrails. Continuous monitoring of model drift (data drift, concept drift, prediction confidence); automated rollback triggers when metrics cross thresholds; human-in-the-loop escalation for decisions above configurable confidence or risk levels; and immutable audit logs that capture inputs, outputs, confidence, and model version for every inference.

03

Lifecycle Governance

Across the model lifecycle: versioned model registry with approval workflows; retirement and sunset policies for deprecated models; regulatory reporting automation (EU AI Act conformity, NIST RMF self-assessments); and periodic third-party audit windows for critical infrastructure models.

PLATFORM-NATIVE, NOT PARALLEL

The key architectural decision: AI governance must be part of the platform, not bolted on. The same platform engineering practices that operators use for their infrastructure—automated CI/CD, immutable infrastructure, observability—must extend to AI model deployment and governance. This is not adding bureaucracy. It is adding instrumentation.

From Framework to Deployment

The Govantis team has been designing and operating governance for the world's most complex multi-stakeholder platforms. Our approach to AI governance draws directly from this experience—not from theory, but from building governance systems that must work across organizational boundaries, regulatory regimes, and technology stacks.

Lessons from platform governance applied to AI:

From Platform to AI

Multi-vendor platform governance requires clear interface boundaries, automated certification gates, and vendor-agnostic compliance layers. AI governance follows the same pattern: the governance layer must be independent of the model provider, the inference infrastructure, and the cloud platform.

We have seen what works: boundary definitions that map to risk domains, automated testing as a release gate (not manual review), and observability that gives operators real-time visibility without requiring AI expertise.

From Theory to Practice

Our work with operators deploying AI in network operations has validated three design principles:

1. Start with the highest-risk decision. Begin governance with the model whose failure causes the most harm. Everything else follows.

2. Automate before you manualize. If a governance check can be automated, it should be. Manual review is reserved for decisions that genuinely require human judgement.

3. Measure governance effectiveness. Track time-to-deployment, incident rate, false positive rate, and audit pass rate. Governance should be a metric, not a department.

> **Key insight:** The operators that succeed with AI governance are not the ones with the largest compliance teams. They are the ones that treat governance as an engineering problem—measurable, automatable, improvable. The same CI/CD discipline that transformed software delivery is now transforming governance.

05 · ACTION

Recommendations for Leadership

AI governance for critical infrastructure cannot be delegated to a single team or purchased as a tool. It must be built into the operating model. Based on our experience across multiple operators and infrastructure domains, we recommend five actions.

01

Conduct an AI Governance Maturity Assessment

Map current AI deployments (production, pilot, planned) against regulatory requirements (EU AI Act, sector regulation) and identify gaps. This is the baseline. **Estimativa Govantis — 4-6 weeks.**

02

Define the AI Governance Operating Model

Who decides what is safe enough? How does AI governance integrate with existing change control, risk management, and compliance functions? Design the RACI, not the tool. **Estimativa Govantis — 6-8 weeks.**

03

Build the Runtime Governance Layer

Instrument model deployments with drift monitoring, automated rollback, and immutable audit logging. This is the foundation of credible AI governance. **Estimativa Govantis — 8-12 weeks per model family.**

04

Pilot with One High-Risk Model

Choose a model with clear failure modes and governance requirements. Run the full lifecycle—design, deployment, monitoring, retirement—with the new governance model. Learn before scaling. **Estimativa Govantis — 8-12 weeks total.**

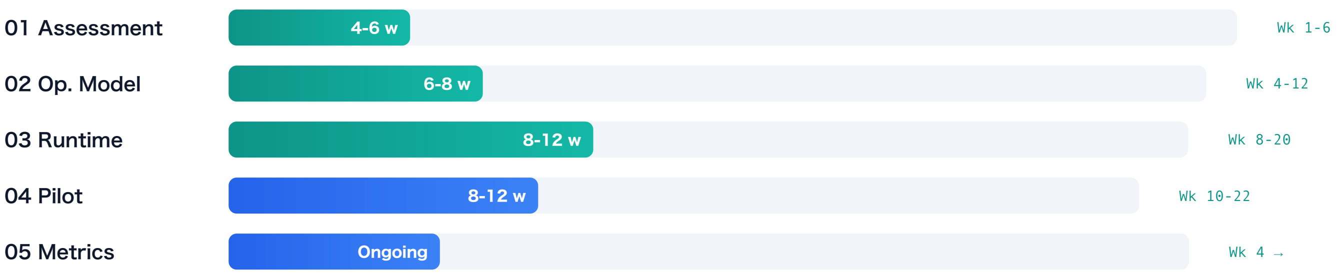
05

Establish Continuous Governance Metrics

Define and track KPIs across deployment velocity, drift detection, and compliance. **Ongoing. First measurable results by week 4.**

VISUAL TIMELINE

Phased execution over 16-22 weeks



CORE METRICS DASHBOARD

Measure what matters

METRIC	WHAT IT TRACKS	TARGET	CADENCE
Model Deployment Velocity	Time from commit to production per model	< 2 weeks	per model
Drift Detection Latency	Time between drift onset and automated alert	< 1 hour	real-time
False Positive Rate	Governance alerts that require no action	< 15%	weekly
Audit Completeness	% of model inferences with full audit trail	99.9%	monthly

Regulatory Compliance Score	% of EU AI Act / NIST RMF requirements met	≥ 90%	quarterly
Human-in-the-Loop Coverage	% of high-risk decisions with proper HITL	100%	weekly

> The cost of waiting for perfect AI governance is the cost of deploying AI without it. Regulators are moving faster than most operators expect.

About Govantis

Govantis is an AI-native platform consultancy founded by engineers who spent decades inside Deutsche Telekom, Vodafone, BT, and NOS. We design architectures, lead platform migrations, and build governance programmes for the world's most complex infrastructure platforms.

Three capabilities. Built for your platform.

Complex Delivery

End-to-end execution on multi-vendor delivery programmes. Broadband, TV, connectivity, and AI infrastructure projects.

AI Products & Scale

Ideate, prototype, and scale AI-native products for complex operational environments. From monitoring to edge intelligence.

Governance & Strategy

Independent governance for complex multi-stakeholder programmes. AI readiness assessment. Regulatory alignment.

Deutsche Telekom

Vodafone

BT

NOS

prpl Foundation

RDK

Let's talk about your platform.

If AI governance for critical infrastructure matters to your business, it matters to us.

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